

THIRD SEMESTER DIPLOMA EXAMINATION IN TEXTILE TECHNOLOGY

DESIGN & STRUCTURE OF FABRICS
(Graph paper should be supplied)

Time : 3 hrs

Maximum marks: 100)

PART - A

(Maximum marks - 10 marks)

Marks

I Answer the following questions in one or two sentences – each question carry 2 marks

1. List the various types of drafts.
2. State what is lifting plan.
3. Distinguish between velvet and velveteen.
4. List the picking order of a double width cloth.
5. Observe the main feature of leno fabric.

PART – B

(Maximum marks = 30)

II. Answer any 5 questions – all questions carry equal marks (5 x 6 = 30)

1. Construct 3x3 mat weave and explain the method of construction.
2. Construct the twill designs with different motifs on four threads.
3. Draw the design for a reversible huck-a-back.
4. Compare warp pile and weft pile structures.
5. Draw the design and thread diagram of tubular cloth.
6. Sketch the thread diagram for a plain pique design on 4 picks.
7. Explain the different methods of producing crepe weaves.

PART – C

(Maximum marks - 60)

(Answer one full question from each module.)

Module I

- III a) Construct a weft combined twill by using 2/2 & 3/3 base twills (8)
b) Draw a 3 x 3 fancy mat weave. (7)

OR

- IV a) Construct a regular Herring bone on 8 threads and explain its method of construction. (8)
b) Draw the plain design and mark the broken draft on 4 healds. Also show the lifting plan, tie-up and treadling plan. (7)

Module II

- V. a) Construct the design and thread diagram for a 5 pick terry and explain the method of formation of pile. (8)
b) Develop a Brighton's honey comb on 12 threads and draw the draft (7)

OR

- VI. a) Construct the thread diagram and design for fast warp pile and explain the method of formation of pile. (8)
b) Draw an ordinary honey comb on 10 threads and explain the method of construction. (7)

Module III

- VII. a) Draw the motif and develop an extra weft design in the ratio of 1:1 (8)
b) Construct the thread diagram and design for a fine cut pique and explain the method. (7)

OR

- VIII a) Develop the design and thread diagram for a wadded Bedford cord with the following ends, Cutting ends - 4 , wadding ends – 4, face ends -12 per repeat. (8)
b) Construct the thread diagram and design for a double width cloth. (7)

Module IV

- IX. a) Draw the design ,draft and denting plan for a 5x5 mock leno. (8)
b) Sketch and describe an open shed formed by using bottom doup heald. (7)

OR

- X. Draw the thread diagram, design and draft of plain gauze structure. (15)